

Children and Climate Change

Climate change impacts on children in the Pacific: A focus on Kiribati and Vanuatu

When extreme conditions become more extreme, how will children fare?

Pacific Islanders have historically managed living with extreme weather. But changing climatic conditions due to human-induced influences appear certain to change their worlds. In fact, changes are already evident. In a world that is largely struggling with rapid urbanization, lack of sanitation infrastructure, scarce water resources and extensive poverty, the changes to come will continue making tough conditions tougher. To that, they will add more turmoil with new challenges.

What are these changes likely to be and how will they affect children? As part of a series of analytical investigations of climate change impacts on children in East Asia and the Pacific, UNICEF looked at the Pacific Islands generally and Kiribati and Vanuatu in particular because their current development challenges are expected to be severely impaired by future global warming.

The Pacific Islands study included climate change modelling by CLIMsystems, a consultant group using internationally recognized state-of-the-art climate change risk and adaptation assessment tools. The modelling was based on two of six potential scenarios that the Intergovernmental Panel on Climate Change (IPCC) has identified as possible future realities due to global greenhouse gas emissions pathways. The A1FI (high population growth and predominant fossil fuel energy sources) and A1B (high population growth and balanced



energy sources) scenarios were applied across an ensemble of 21 climate models. The results support previous IPCC projections that indicate both Kiribati and Vanuatu are likely to see reasonable increases in average temperature as well as considerable changes to the return rate and magnification of extreme temperatures and rainfall events. The results of this mean considerable challenges for food security, economic prosperity, major population displacement and a decreased ability to meet the Millennium Development Goals.

Given the uncertainty associated with the modelling due to limited baseline studies and robust local climate data, it is difficult to clearly identify the geographic and temporal spread of changes to average rainfall. This uncertainty makes it difficult to further explore downscaled local challenges. The data does show that, in general, Kiribati and Vanuatu are likely to experience increased average



temperatures and more frequent occurrences of extreme heat events. For rainfall, the model results shows a wider range of possibilities (both increased and decreased rainfall), with the likelihood that extreme precipitation events will be more frequent. The analysis also identified possible increases in cyclone intensity for Vanuatu and potential for the loss of the majority of landmass for Kiribati (due to sea level rise). For a more comprehensive overview of the projected scenarios and impacts, refer to the comprehensive technical report (Burton et al., 2011).

What this means for children is an increase in the exposure to a more extreme climate and all the associated implications to their basic rights to family, shelter, nutritious food, health care, education, hygiene and protection from abuse.

Given that children are still developing and that their future productivity (and thus the development prospects of any one country) can be impaired by detriments to any of their basic rights, they are essentially looking at climate change challenges from a more vulnerable position. And given that young

people make up 57 per cent of the Pacific population (37 per cent of them younger than 15 years and 20 per cent aged 15–24) (Larsen, 2011; UNICEF, 2010), children's vulnerability makes countries even more vulnerable to the negative side of the impacts. The social protection burden on governments and civil society alone will likely increase tremendously.

How big will be that burden can, to a degree, be controlled by how well governments and civil society can adapt to and prepare for anticipated impacts and thus pre-empt or diminish their severity. The Pacific situation is not beyond mitigation responses. But it is not beyond the possibility for positive change that can offset the coming negative changes. As the comprehensive analysis report points out, considerable opportunities exist for children-focused and children-led responses to the effects of climate change impacts.

The Pacific Island context

The Pacific region consists mainly of Small Island Developing States, which without doubt are particularly vulnerable to the

climate change impacts due to such physical characteristics as low-lying topography and dependence on natural resources. Additionally, a large proportion of countries in the Pacific have complex social, environmental, economic and political challenges that may weaken their responses to climate change. This is in part due to rapid population growth, urbanization, geographic isolation, international lobbying power and other issues associated with historical sovereignty.

What the study looked at in Kiribati and Vanuatu

In looking at potential impacts of climate change on children in Kiribati and Vanuatu, the Pacific study examined four primary elements, the analysis of which is highlighted in this briefing and elaborated on in the comprehensive technical report:

- temperature
- precipitation
- sea level rise
- climate change policies

Temperature-related impacts

In Kiribati and Vanuatu, the average maximum temperatures are expected to increase by 2050 to just below the projected global average (of 2.7°C), with the results of the modelling showing an increase of approximately 2°C relative to the 1960–1990 average. Extreme temperature events (such as heat waves) can impact on the health and well-being of children; studies show a correlation between ambient temperature and diarrhoea-related hospitalizations for children (between 1 and 5 per cent increase per 1°C of warming) (Kolstad and Johansson, 2011). Empirical studies also highlight a correlation between extreme temperatures and increased violence in developed nations, such as the United States (Doherty and Clayton, 2011) as well as reports of increasing occurrence of domestic violence immediately after natural disasters in the Pacific (Kingi, 2011). Domestic violence is an existing widespread problem in Kiribati (Kingi and Roguski, 2011) and Vanuatu (Fairbairn-Dunlop, 2009).

Geographic temperature variations exist (especially in the higher altitudes in Vanuatu), meaning that heat-related impacts will vary depending on location. For example, densely populated squatter settlements in Port Vila,

Vanuatu, are likely to be more at risk from extreme heat events than villages higher in elevation and surrounded by vegetation (such as villages in Santo).

Historically, the number of tropical cyclones in Vanuatu has increased. An escalation of the global average temperatures and sea surface temperatures may result in more intense tropical cyclones, which are likely to result in a cascade of issues for children, including physical injury to them or their families, loss of shelter, reduced access to education and health facilities as well as post-disaster psychological issues. For example, Tropical Cyclone Atu (February 2011) resulted in almost 3,000 families (or 6 per cent of Vanuatu's population) requiring aid (including food, water and sanitation support) (Red Cross, 2011).

Impacts associated with rainfall

Both countries are likely to experience periods of reduced rainfall and increased rainfall (longer drier periods followed by increased rainfall intensity). Both countries have variable rainfall across their broad geographic expanse and the modelling suggests that the variability will continue.

Decreases in average rainfall reduce the ability for water catchment (increasing the reliance on dwindling freshwater lenses). Simpson et al. (2009: 183) shows that "a 10 per cent reduction in average rainfall by 2050 is likely to correspond to a 20 per cent reduction in the size of the freshwater lens on Tarawa atoll, Kiribati." Combined with an increase in sea level rise and increased population growth, the current problem with safe water scarcity will be magnified considerably, and along with it the implications for children's health and parents' productivity (which falls with chronic diarrhoea-related illness (UNICEF, 2008).

Supply of potable water is also an issue in Vanuatu, although it has greater water availability potential than Kiribati. In some islands and squatter communities, however, access to drinking water is problematic due to the lack of basic infrastructure and limited availability and quality of freshwater lenses. As fishing livelihoods diminish due to increased sea surface temperatures, ocean acidification and coastal erosion, more families are expected to move to cities and most likely to existing or new squatter communities.



Reduced resources could decrease children's opportunities for education

In Vanuatu, decreased rainfall will likely impact on livestock and crops, resulting in economic and food (including nutrient) security issues for children and their families (Reti, 2007). According to the World Bank (2011), already more than 11 per cent of children younger than 5 years in Vanuatu suffer from malnutrition.

Increases in rainfall may help support rainwater harvesting (depending on the size of the water tank). It also can have negative impacts, including increased runoff-related issues for coral reefs, localized flooding, landslides (in Vanuatu) and crop damage (Reti, 2007).

Impacts associated with sea level rise

Climate change-related issues enveloping Small Island Developing States, such as Kiribati, are dominated by the projections of sea level rise. The low end of the projections will still require considerable roll-out of infrastructure solutions (such as sea walls and water-storage facilities). In a best-case scenario, it is more than likely that some communities in both Kiribati and Vanuatu will need to relocate to less exposed areas.

The geographic spread of sea level rise-related risks for children in Kiribati is ubiquitous. By 2050 it is likely that sea level rise will have direct and indirect consequences for all, due to the nation's low-lying nature. The incremental process of sea level rise will at first affect those who do not have sufficient capacity (or space) to avoid the risk. For example, in Betio, the density of the 12,500 population is likely to inhibit options for moving from the

coastal foreshore. Some communities (like Tebunginako in Abaiang province, with a population of more than 300) already have had to relocate as a result of coastal erosion over the past few decades (Reed, 2011).

In Kiribati, the majority of children's homes, health facilities, schools, churches and recreation areas are located within a few hundred metres of the coast. A mapping of sea level rise impacts (for the year 2100) on parts of the Kiribati capital of Tarawa (an atoll), under an IPCC high-impact scenario (A1FI) indicate at least six south-side villages that are threatened with high or extreme risks). The risks presented include coastal erosion, increased exposure to high-tide events, inundation of the freshwater lens, increased infrastructure maintenance requirements and the potential for land to become uninhabitable (Elrick, Kay and Bond, 2009).

Going by those high-end sea level rise projections, the very existence of the country is in jeopardy. A World Bank report in 2000 projected that by 2050 "in years of strong storm surge, up to 54 per cent of South Tarawa could be inundated, with capital losses of up to US\$430 million." That projection used a sea level rise range of 23–43 centimetres. Analysing graphs by Vermeer and Rahmstorf (2010) indicate a rise that is between 30 and 50 centimetres.

For Vanuatu, sea level rise initially presents a direct threat to those in low-lying coastal areas (although any impact on coastal resources is likely to affect most communities due to knock-on effects). An entire village in the Tegua Island community in Torba province (north Vanuatu) recently relocated due to increased coastal erosion (due to a combination of seismic land movement and sea level rise). Sea level rise in Vanuatu is difficult to estimate due to the frequent seismic activity, with some coastal areas experiencing both upward and downward land movements.

The most pressing challenges for children from sea level rise in Kiribati and Vanuatu are likely to include:

- psychological issues associated with loss of familiar surroundings and adjustment into new settlements
- dwindling portable water supplies as freshwater lenses decrease

- increased risk of diarrhoea and its consequences from the spread of faecal matter due to inadequate sanitation in overcrowded settlements (especially in Kiribati urbanized areas)
- diminished access to education and health facilities during increasing king tide events or other extreme weather events (such as cyclones).

Impacts associated with climate change policies

As well as the biophysical risks, there is potential for children to be affected by climate change adaptation and mitigation policies. Although policies are created with the best intentions, there is always the risk that they can lead to unexpected negative impacts. This is especially so if there has been little engagement with communities and children or little integrated planning (of many issues). For example, the 'safer islands' approach is a suggested adaptation regime being considered in Pacific countries in which certain population centres are prioritized in terms of investments in infrastructure and services.

Under the safer islands approach, the selected areas become relatively more resilient to climate impacts with increased support but are then exposed to a potential range of unexpected issues. The increase in more resilient services and economic development may lead to these areas experiencing disproportionate urban growth. In South Tarawa, Kiribati, any further rapid growth is likely to result in considerable challenges for children and the wider community, given the already limited space and constraints on the resources. For example, dense settlements create an urban heat island effect and increased population will exert increased pressures for proper and safe water supply and sanitation.

In addition, there likely will be an increase in children temporarily relocating from their homes in outer islands to the newly built-up areas for their education. Among the risks, documented experiences indicate that children who are billeted out to extended families face an increased threat of abuse (United Nations ESCAP, 2008).

Adaptation plans may impose psychological stress on children. According to Doherty and Clayton (2011), "Forced relocations can involve a severing of emotional ties to place,

disruption of existing social networks, and attempts to maintain cultural integrity despite relocation... These disruptions of geographic and social connections may lead to grief, anxiety, and a sense of loss, particularly among those with a strong [sense of] place or national identity."

There is no evidence showing that psychological issues associated with planned retreat (inter- and intra-country) for children are being considered in Kiribati or Vanuatu. Relocation in parts of Kiribati and Vanuatu has already occurred, but there is no literature that discusses the ramifications that these events may have, or are having, on children. The relocation is necessary, but the plans need to include services for helping children cope. Even an awareness of the possibility of relocation or of future climate risks has the potential to affect a person's well-being by manifesting as anxiety, grief or apathy (Doherty and Clayton, 2011).

As the impacts of sea level rise become more evident globally, Kiribati and Vanuatu will no doubt be competing for funds with other low-lying nations (including those where millions of people are exposed to the impacts, including Bangladesh and Egypt. This presents resource limitations on the adaptation options available. It is becoming increasingly evident that a global triage approach to managing sea level rise risks may occur, with a focus on funding and infrastructure investments placed on the most populated areas first. For children, this may mean having to migrate to more resilient locations that provide education and health services.

Other challenges presented by adaptation plans include international migration, which can result in cultural loss, loss of ecological knowledge, sense of belonging and psychological impacts (Adger et al., 2011); seawalls can affect access to traditional fishing locations (Damodaran, 2003); poor rainwater tank installation can harbour mosquitoes, which can result in increased malaria (in Vanuatu) and/or dengue fever (Vanuatu and Kiribati; Mariappan, 2007); and a global carbon-constrained economy may present initial increased costs for energy and transport, which will consume a larger proportion of poor household income, thus possibly impacting nutritious food consumption, health care or education.

Climate change adaptation and mitigation policies of course are absolutely necessary. Without considerable reductions in anthropogenic greenhouse gas emissions, mass social, environmental and economic upheaval is likely. Adaptation has become a development issue and is already needed to provide increased resilience for vulnerable communities in terms of increased safe water and food security, improved access to health care and social services, and directing economic investment towards low-carbon alternatives and away from high-risk areas.

Considering children in Pacific adaptation plans

Across the 14 Pacific countries' adaptation plans reviewed for this study, children's needs are rarely mentioned in the existing climate change-related documents. Most Pacific nations are developing climate change risk assessments and adaptation plans. With many of the Pacific countries having least developed country status, their climate change adaptation planning processes tend to follow the United Nations Framework Convention on Climate Change (UNFCCC), which provides funding for national adaptation programmes of action (NAPA). But the NAPA guidelines do not mention children's rights or have specific provision for their needs. Most countries in the Pacific have committed to the Pacific Islands Framework for Action on Climate Change 2006–2015; it identifies regional opportunities for implementing adaptation, networking, shared learning and common goals, but, as well, there is no specific mention of children.

Most of the NAPAs or national communications (the country-specific reports to the UNFCCC) in the Pacific region¹ acknowledge that children are particularly vulnerable to the impacts (especially their health). Only Kiribati and the Solomon Islands, however, clearly indicate a pathway for including children's views in their respective adaptation processes (Burton et al., 2011).

Vanuatu is among the few Pacific countries that have both a NAPA and a disaster risk reduction

programme. A recent UNICEF review (2009) of its emergency preparedness plan indicates that Core Commitments for Children indicators were not adequately considered (such as child separation, child protection or temporary classrooms in emergency management plans). Vanuatu's NAPA contains no mention of youth or children associated with climate change other than considerations in a food security study.

Disaster risk reduction components are being incorporated into the Kiribati Adaptation Plan, which emphasizes the inclusion of children in all decision-making processes (especially during relocation discussions).² As in Vanuatu, the UNICEF review of emergency planning in Kiribati against the Core Commitments for Children indicators also found a less-than-satisfactory score.

Opportunities for action

Identifying the specific challenges from biophysical climate change risks is a complex task. The uncertainties in models and responses of the natural environment mean that an adaptive management process is critical for exploring the potential risks. Typically, children are not involved in decision-making processes; their well-being and future is directly related to what decisions their parents and policy-makers make and how aware they are about climate change impacts and environmental change. Although, increasing awareness of climate change does not necessarily result in more action or preparedness if it is not simultaneously matched with increases in capacities and resources to adapt.

The following recommendations highlight the opportunities for the inclusion and consideration of children in climate change adaptation.

Recommendations

Include children's views

Including the opinions of children (and youth) should start with specific recognition within

¹ Due to the scope and funding limitations of this UNICEF climate change impact study, only the NAPAs and National Communications to the UNFCCC were reviewed. A comprehensive review of all Pacific adaptation actions (NGO activities, regional partnerships) is needed to gain a more comprehensive assessment of how children's needs are integrated or not into adaptation plans.

² The degree of 'inclusion' of children in the process is not clear and will become more evident after the third phase begins in July 2011.

Children help to protect marine ecosystems

Coastal erosion amplified by sea level rise is an ongoing concern for many communities in Kiribati. One response to tackle this problem is to use sea walls in vulnerable locations. However, sea walls are expensive to build and have some negative side effects (reduced access to the shoreline and impacts on natural coastal systems). Erosion can also be managed by using softer approaches, such as planting trees. The Kiribati Government recently planted more than 37,000 mangroves as part of its adaptation project because of their high efficiency in protecting coastlines from erosion. They also have a wide range of ancillary benefits, such as nurseries for fish, filtering water runoff and sequestering carbon dioxide.

Many children are now helping to plant mangroves. According to Turang Faveae, Biodiversity and Conservation Officer at the Environment and Conservation Division of the Ministry of Environment, communities have asked to be a part of the mangrove-planting projects. For example, in the small islet community of Takaeang in Aranuka atoll, the community had concerns about coastal erosion threatening their school and asked to be included in the mangrove planting programme. Students at Takaeang primary school helped plant 308 mangroves and in the process learned about the challenges of climate change. “We encourage and practise mangrove planting with communities, youth groups and school students so they can see the importance of planting and gain a sense of ownership to look after and manage the mangroves,” explained Turang Faveae (World Bank, 2011). Officers with the Environment and Conservation Division regularly visit schools to teach children about the natural environment and climate change.

In Vanuatu, children are engaging in river clean-up projects, which are used also to increase their environmental awareness. Additionally, schools participate in environmental programmes such as Reef Check, in which students visit coral reefs and learn about the marine ecosystems and what needs to be done to protect them. The role of education is recognized as an important channel of climate change and adaptation awareness, which is why out-of-school children are cut off from information that is vital for their present and future lives – their access to information needs to be addressed. Adaptation actions that are practical and have an educational component are important in raising climate awareness.



the NAPA guidelines. This can be done through requirements in donor funding agreements for children’s participation and consideration (prescriptive proportions in meetings and summaries for children in all climate change reports).

Involving children and youth in policy- and decision-making processes simultaneously becomes an investment in their capacity building as future leaders and better-informed citizens.

Get the policies right

Even with best intentions, adaptation actions have the potential to negatively impact on the health and well-being of children. For

example, even if relocation is required, impacts associated with dislocation are likely to emerge. Thus, a balance needs to be forged between providing technological solutions for physical impacts and increasing the adaptive and coping capacities of individuals and communities who will be most affected by climate change (such as research to identify psychological pressure points associated with relocation; providing processes that maintain connections to land and culture once relocation is eminent; and guiding principles for assessing potential impacts that climate change policies may have on children). Adaptive management is needed to ensure that the policies maintain no-impact or low-



Involving children in climate change discussions provides important opportunities for more inclusive policy-making

impact solutions. This can be achieved by building in indicators that include monitoring and evaluation systems to know when policies might need to change. Involving children in long-term visions of what the future should and could look like are also crucial to get the policies right.

Teach sensitively

Children are sensitive to discussions surrounding negative environmental impacts. Emphasis should be given in education programmes to avoid 'doom and gloom' scenarios and instead provide children with information on positive options and how they can help. A balance between children's psychological well-being and their right to information needs to be managed accordingly: this means being mindful of possible psychological stresses created by frightful information regarding climate change impacts.

Incorporate active learning into children's activities that allow them to 'learn by doing'. For example, installing low-cost weather stations in schools and/or in villages may help foster children's awareness of how the weather and climate may affect them and their community. Children who have been educated on how to record climate data can share and compare their results with other schools, including overseas interaction.

Education about climate change for children needs to be country specific and use children's own language, such as Bislama (and other local languages) in Vanuatu and Kiribati in Kiribati. Incorporating climate change into existing environmental awareness programmes, courses

and workshops is an option that has been trialled in both countries. For instance, some NGOs have included climate change as part of their environmental awareness education programmes, and both the Kiribati and Vanuatu governments have conducted workshops integrating issues concerning the environment and climate adaptation. Mainstreaming climate change rather than treating it as a separate issue is essential, given the already constrained resource base and the issues these countries already face.

Children also need to be given information that is in an easily understandable format. The majority of climate change reports are complex. For example, the IPCC reports are synthesized into a summary for policy-makers to ensure that the language suits that particular audience. It is important that climate change reports (IPCC documents, country NAPA reports, etc.) contain a 'summary for children' component.

Gather information

Reviewing climate science regularly is necessary for any decision-making, particularly given the changing projections on sea level rise and extreme weather events. Collecting data also on non-physical elements, such as the number of saline wells, erosion events leading to partial relocation of infrastructure and people and the number of times children miss school because of extreme events, are important indicators for monitoring societal and environmental change. Communities can be integral in providing and documenting such data, which can also inform the people responsible for making decisions and policies on climate adaptation.

Share information

Given that funding for climate adaptation has increased rapidly in the past few years in the Pacific, information sharing is critical. Both Vanuatu and Kiribati should invest in information-sharing mechanisms between donors, NGOs, governments and other actors to both learn from successes and increase the deliverability of adaptation action on the ground. One option is to collate databases so that information is widely accessible on the latest climate change science, climate-specific projects, funding arrangements and various people's needs for collaboration. Simply knowing who does what, where, when and

how will enable more efficient approaches within both countries. Identifying what impacts projects and programmes can have on children is essential to incorporate child-sensitive approaches.

This could be done, for example, by establishing specific child-focused indicators within projects and programmes that would be used to collect information on, such as:

- number of relocated children due to environmental change (including flooding, erosion, extreme events)
- number of children involved in and accessing benefits from food security projects
- number of children accessing new water infrastructure (including tanks and potable water)
- number of children involved in climate adaptation activities in schools and groups
- number or proportion of children included in meetings for adaptation planning.

Collaborative workshops are needed to allow for the strengthening of networks and to disseminate the latest country-specific climate science. Examples of useful collaborative workshops include sector-specific knowledge sharing (all health departments in each Pacific nation meet to share methods, research and challenges identified in managing health effects that climate change may have on children).

Address the threats to children's well-being

In the Pacific, children face several threats to their health status and well-being, such as lack of access to safe drinking water, lack of access to education, low household income and limited access to health facilities. These threats are very likely to be exacerbated by climate change impacts. Donors, NGOs and governments should consider their policies and programmes from a child-related perspective, particularly when forming climate adaptation responses. Investing, for instance, in increased drinking water availability and quality in schools, as the Kiribati Adaptation Project is doing, is important in delivering development and adaptation goals and will have direct effect on children's well-being.

Understand the links between migration and food shortages

Investing in research, policies and programmes regarding the role of migration and food shortage will help identify the factors that



Increased understanding of the links between food security, climate change and migration is crucial for increased resilience and effective responses

Delivering basic development outcomes, such as increased water availability and quality, can make a crucial difference for children's well-being



influence relocation strategies among children and youth in the Pacific. Both Vanuatu and Kiribati have projects looking at climate change and food security issues, but these may not necessarily address the needs and vulnerabilities of children in particular. Given that many children in both countries relocate to urban centres for education and employment opportunities, it is important that donors, NGOs and public institutions specifically pay attention to how these groups are dealt with, what services are available and how the vulnerabilities specific to them may be exacerbated by climate change.

Conclusion

There is little doubt that Small Island Developing States will face considerable challenges arising from human-induced

climate change, with children at the forefront of these impacts. The degree of impact from global warming on their nations will be determined by the rest of the international community's ability and willingness to reduce greenhouse gas emissions. The Small Island Developing States in the Pacific are geographically and economically vulnerable to the impacts of climate change, which are likely to emerge sooner rather than later.

Although often constrained economically, the children of Pacific Island nations have a wealth of cultural practices and close family networks that they can draw on to contribute to the adaptation needs of their communities and countries. Well-planned actions at the government and donor levels are required to enable those contributions.

Agent/actor	Opportunities for adaptive action and capacity building
Government	• Embed children's rights into all climate change policies and planning • Ensure all climate change policies contain a 'summary for children' chapter • Undertake sensitive climate change education campaigns • Create a framework for assessing climate policies against the impacts on children's rights • Undertake studies that explore the impacts of relocation • Identify and support sentinel sites (in conjunction with UNICEF) for early indicators of climate-related impacts
Children	• Disseminate information about climate change impacts and adaptation (take information learned at school and help increase awareness at home) • Engage in soft infrastructure solutions (such as mangrove planting) • Participate in the planning for extreme weather and climate change (included in actor discussions) • Use active learning to help inform about climate change and extreme weather (local weather stations) • Share their stories about impacts and adaptation actions with national and international audiences via Internet and /or conferences • Participate in creating a framework for summary for children chapters – including advice on good information dissemination platforms (fact sheets, online, mobile phone, social media, school, theatre, etc.)
Development partners	• Ensure that all projects have children's rights embedded into the objectives • Include a summary for children (similar to summary for policy-makers) in all project reports • Work with countries and UN agencies to develop guiding principles for consideration of children in adaptation planning
UN agencies	• Embed considerations for children's rights in guidelines for the UNFCCC NAPAs and any future adaptation mechanisms • Create an IPCC summary for children chapter for all IPCC reports (similar to summary for policy-makers) • Create a framework that allows for the evaluation of proposed adaptation actions and projects to assess potential impacts on children's rights • Undertake child-specific impact analysis for adaptation actions and guidelines • Create guidelines for the summary for children and summary for child advocacy

References

- Adger, W. N., Barnett, J., Chapin, F. S. and Ellemor, H., 'This Must Be the Place: Underrepresentation of Identity and Meaning in Climate Change Decision-Making' in *Global Environmental Politics*, 2011: 11, 1-25.
- Burton, D., Mustelin, J., and Urich, P., *Climate Change Impacts on Children in the Pacific: Kiribati and Vanuatu: Technical report*, commissioned by UNICEF (forthcoming), 2011.
- Damodaran, A. M., 'Property Rights, Management Plans and Dynamics of Sustainability in the Coastal Kerala', in *Multidisciplinary Scientific Conference on Sustainable Coastal Zone Management* 12-14 June 2003, Stockholm, 2003, available at: www.hutten.org/fw/docs/304.pdf.
- Doherty, T. J. and Clayton, S. D., 'The Psychological Impacts of Global Climate Change', in *American Psychologist*, 2011: 66, 265-276.
- Elrick, C., Kay, R. and Bond, T., *Planning Manual: Supporting land use decision making in the Republic of Kiribati*, Prepared for Kiribati Adaptation Project Phase II (KAP II), Government of Kiribati, 2009, available at: www.climate.gov.ki/pdf/Adaptation/KAP%20Planning%20Manual.pdf.
- Kolstad, E. and Johansson, K., 'Uncertainties Associated with Quantifying Climate Change Impacts on Human Health: A case study for diarrhea', in *Environmental Health Perspectives*, 2011: 119 (3): 299-305.
- Larsen, J., 'The trafficking of children in the Asia-Pacific', Australian Institute of Criminology, April 2011 (Canberra), 2011, available at: www.aic.gov.au/publications/current%20series/tandi/401-420/tandi415.aspx
- Mariappan, T., 'Monitoring of Rainwater Harvesting System Vital to Avoid Proliferation of Mosquito Vectors', in *Current Science*, 2007: 92: 12.
- Red Cross 2011, DREF operation update: Vanuatu Tropical Cyclone Vania, 2011, available at: www.ifrc.org/docs/appeals/11/MDRVU001drefOU1.pdf
- Reed, B., 'Climate Change and Faith Collide in Kiribati', National Public Radio, 2011, available at: www.npr.org/2011/02/16/133650679/climate-change-and-faith-collide-in-kiribati?ft=3&f=133681251.
- Reti, J. 2007. *An Assessment of the Impact of Climate Change on Agriculture and Food Security in the Pacific: A case study in Vanuatu*. Pacific Environment Consultants Ltd (PECL).
- Simpson, M. C., Scott, D., New, M., Sim, R., Smith, D., Harrison, M., Eakin, C. M., Warrick, R., Strong, A. E., Kouwenhoven, P., Harrison, S., WILSON, M., Nelson, G. C., Donner, S., Kay, R., Geldhill, D. K., Liu, G., Morgan, J. A., Kleypas, J. A., Mumby, P. J., Christensen, T. R. L., Baskett, M. L., Skirving, W. J., Elrick, C., Taylor, M., Bell, J., Rutt, M., Burnett, J. B., Overmars, M., Robertson, R. and Stager, H., *An Overview of Modeling Climate Change Impacts in the Caribbean Region with Contribution from the Pacific Islands*, United Nations Development Programme (UNDP), Barbados, West Indies, 2009.
- United Nations Children's Fund (UNICEF), *Promotion of Household Water Treatment and Safe Storage in UNICEF WASH Programs*, 2008, available at: www.unicef.org/wash/files/Scaling_up_HWTS_Jan_25th_with_comments.pdf
- UNICEF, *Think children! Using the Core Commitments for Children to review government emergency preparedness and response plans and capacity*, 2009, available at: www.unicef.org/pacificislands/Think_children_inside_pages_FINAL_24.04.09.pdf
- United Nations ESCAP, *Regional Stakeholders' Consultation and Planning on the Commercial Sexual Exploitation of Children and Child Sex Abuse in the Pacific*, 2007, available at: www.unescap.org/ESID/GAD/Events/RegionalConsultationNov07/ESCAP_CSEC_Pacific-Regional-Consultation-report_2007.pdf

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